

EXPERIMENT– 2 Calculator

PROJECT NAME

Simple Calculator Using Arduino UNO, 4x4 Keypad and SSD1306 OLED Display

OBJECTIVE / PROBLEM STATEMENT

This project implements a simple calculator using Arduino UNO, a 4x4 keypad, and an SSD1306 OLED display. The calculator performs basic arithmetic operations such as addition, subtraction, multiplication, and division. The result is displayed on the OLED screen.

COMPONENTS USED

Component	Quantity
Arduino UNO	1
4x4 Keypad	1
SSD1306 OLED Display	1
Jumper Wires	Required
Breadboard	1

PIN CONNECTIONS

4x4 KEYPAD CONNECTIONS

Keypad Pin	Arduino Pin
R1	D1
R2	D2
R3	D3
R4	D4
C1	D5
C2	D6
C3	D7
C4	D8

Symbols mapped operators

Symbol	Operation
A	+
B	-
C	*
D	/
#	AC

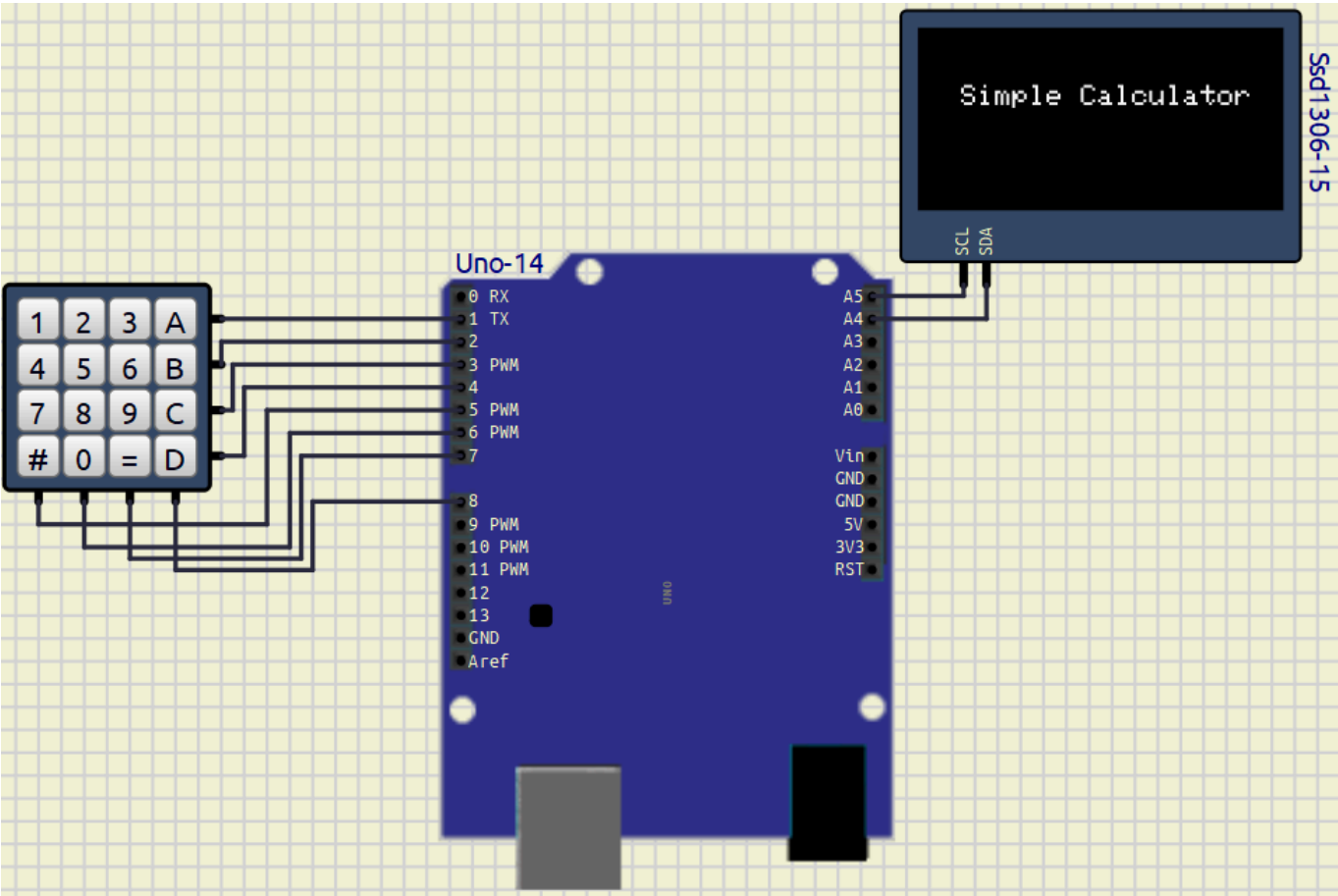
SSD1306 OLED DISPLAY CONNECTIONS

OLED Pin	Arduino Pin
VCC	5V
GND	GND
SDA	A4
SCL	A5

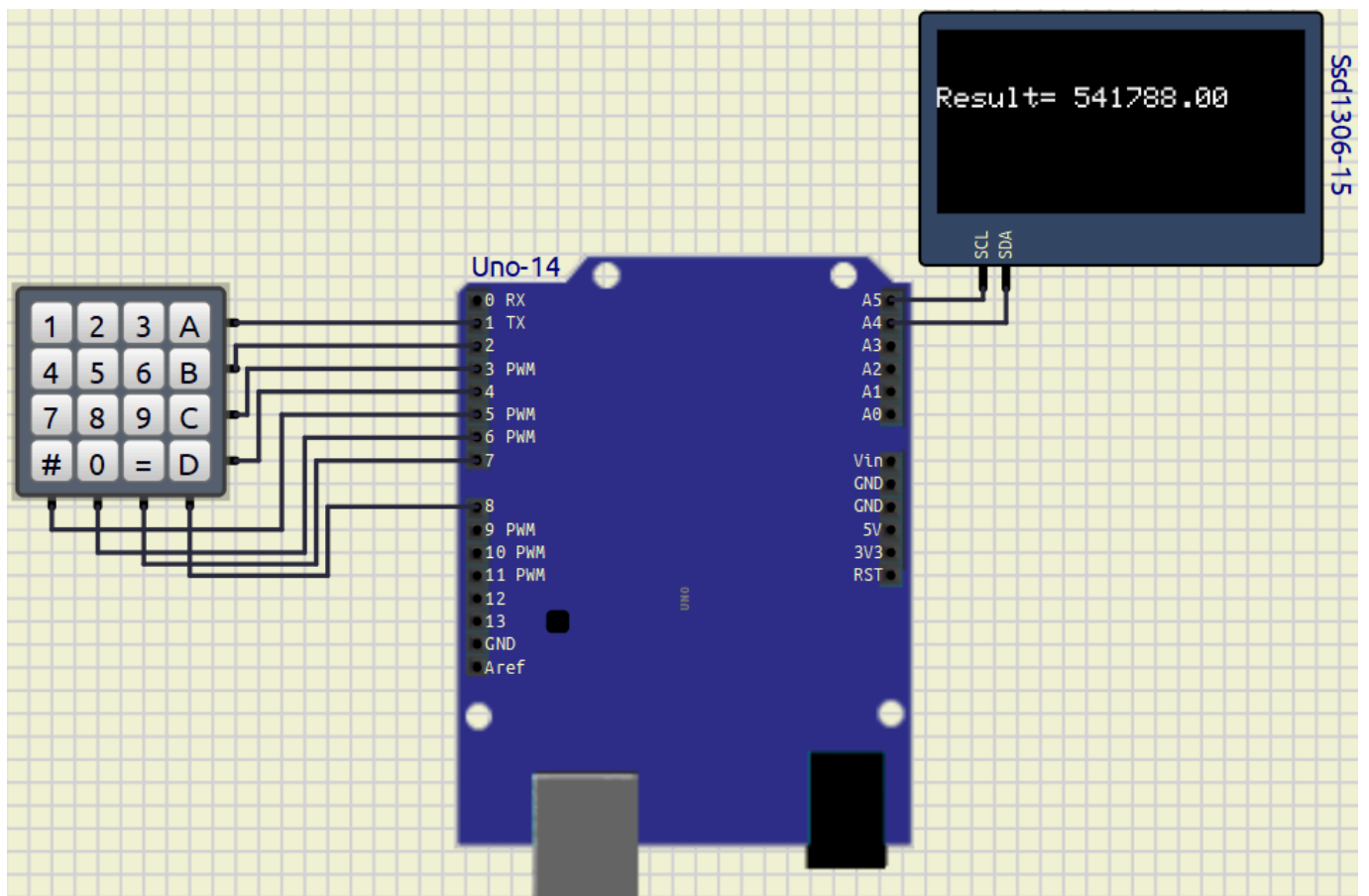
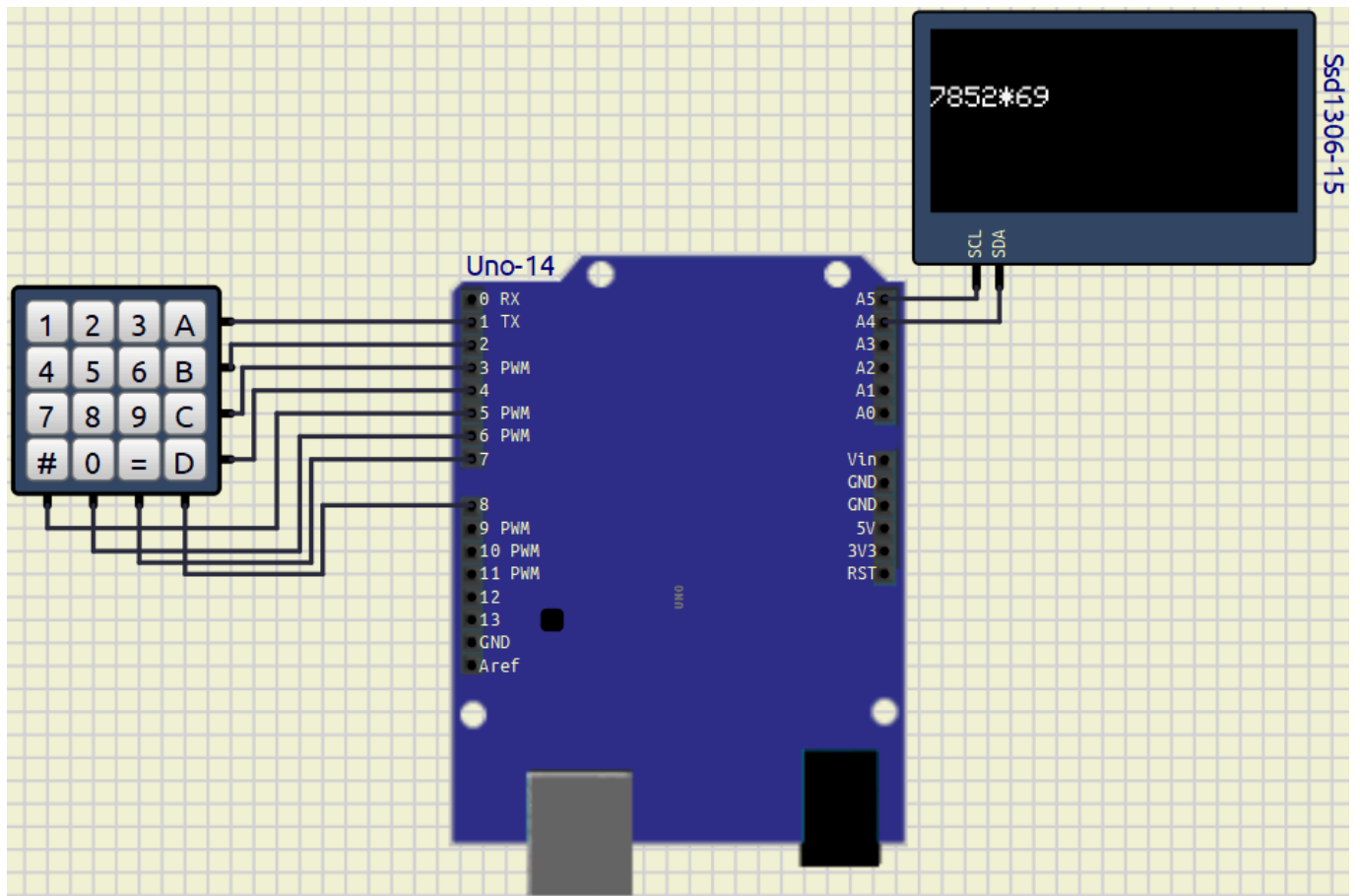
SOFTWARE USED

- Arduino IDE
 - SimulIDE
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CIRCUIT DIAGRAM



Example:



REQUIRED LIBRARIES

```
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#include <Keypad.h>
```

MAIN ARDUINO CODE

Code With Comments ---> [Click here](#)

```
// =====
// SIMPLE CALCULATOR USING:
// 1. Arduino UNO
// 2. 4x4 Keypad
// 3. SSD1306 OLED Display
// =====

// ----- LIBRARIES -----
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#include <Keypad.h>

// ----- OLED SETTINGS -----
#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64

// Create OLED display object
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);

// ----- KEYPAD SETTINGS -----

// Keypad size
const byte ROWS = 4;
const byte COLS = 4;

// Keypad layout
char keys[ROWS][COLS] = {
  {'1', '2', '3', '+'},
  {'4', '5', '6', '-'},
  {'7', '8', '9', '*'},
  {'#', '0', '=', '/'}
};

// Arduino pins connected to keypad
byte rowPins[ROWS] = {1, 2, 3, 4}; // R1 R2 R3 R4
byte colPins[COLS] = {5, 6, 7, 8}; // C1 C2 C3 C4

// Create keypad object
Keypad keypad = Keypad(makeKeymap(keys),
```

```
        rowPins,
        colPins,
        ROWS,
        COLS);

// ----- VARIABLES -----

// First number
String num1 = "";

// Second number
String num2 = "";

// Operator
char op;

// Result
float result;

// True after operator pressed
bool operatorPressed = false;

// Full expression for display
String expression = "";

// =====
// SETUP
// =====
void setup() {

    // Start OLED display
    display.begin(SSD1306_SWITCHCAPVCC, 0x3C);

    // Clear screen
    display.clearDisplay();

    // Text settings
    display.setTextSize(1.8);
    display.setTextColor(WHITE);

    // Welcome message
    display.setCursor(15, 20);
    display.println("Simple Calculator");

    display.display();

    delay(2000);

    display.clearDisplay();
    display.display();
}

// =====
// LOOP
```

```
// =====  
void loop() {  
  
    // Read pressed key  
    char key = keypad.getKey();  
  
    // If key pressed  
    if (key) {  
  
        // =====  
        // CLEAR EVERYTHING  
        // =====  
        if (key == '#') {  
  
            num1 = "";  
            num2 = "";  
            expression = "";  
            operatorPressed = false;  
            result = 0;  
  
            display.clearDisplay();  
            display.display();  
        }  
  
        // =====  
        // IF NUMBER PRESSED  
        // =====  
        else if (isDigit(key)) {  
  
            // Store first number  
            if (!operatorPressed) {  
                num1 += key;  
            }  
  
            // Store second number  
            else {  
                num2 += key;  
            }  
  
            // Add to expression  
            expression += key;  
  
            // Show on display  
            updateDisplay(expression);  
        }  
  
        // =====  
        // IF OPERATOR PRESSED  
        // =====  
        else if (key == '+' || key == '-' ||  
                key == '*' || key == '/') {  
  
            // Allow only one operator  
            if (!operatorPressed && num1 != "") {
```

```
        op = key;
        operatorPressed = true;

        expression += key;

        updateDisplay(expression);
    }
}

// =====
// CALCULATE WHEN '=' PRESSED
// =====
else if (key == '=') {

    // Convert strings to numbers
    float n1 = num1.toFloat();
    float n2 = num2.toFloat();

    // Perform calculation
    switch (op) {

        case '+':
            result = n1 + n2;
            break;

        case '-':
            result = n1 - n2;
            break;

        case '*':
            result = n1 * n2;
            break;

        case '/':

            // Prevent divide by zero
            if (n2 != 0) {
                result = n1 / n2;
            }
            else {

                display.clearDisplay();

                display.setCursor(0, 20);
                display.setTextSize(1);
                display.println("ERROR");

                display.display();

                delay(2000);

                return;
            }
    }
}
```



```

        break;
    }

    // Final expression
    expression = "Result= " + String(result);
    // Show result
    updateDisplay(expression);

    // Prepare for next calculation
    num1 = String(result);
    num2 = "";
    operatorPressed = false;

    // Keep only result for next operation
    expression = String(result);
}
}
}

// =====
// DISPLAY FUNCTION
// =====
void updateDisplay(String text) {

    display.clearDisplay();

    display.setTextSize(1);
    display.setCursor(0, 20);

    display.println(text);

    display.display();
}

```

WORKING PRINCIPLE

1. The 4x4 keypad is used to enter numbers and operators.
 2. Arduino UNO continuously scans the keypad.
 3. Entered values are stored as operands.
 4. The selected operator determines the operation to be performed.
 5. When '=' is pressed:
 - Arduino performs the calculation.
 - Result is displayed on the SSD1306 OLED display.
 6. '#' key clears all previous inputs.
 7. The process repeats continuously.
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OUTPUT

Startup Screen

Simple Calculator

Example Calculation

5+3

Result= 8

FEATURES

- Performs Addition
 - Performs Subtraction
 - Performs Multiplication
 - Performs Division
 - OLED Display Output
 - Clear Function using '#'
 - Real-time Input Detection
 - Divide by Zero Error Handling
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ADVANTAGES

- Simple and easy to implement
 - Low-cost embedded project
 - Real-time calculations
 - User-friendly interface
 - Useful for learning keypad interfacing
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APPLICATIONS

- Basic calculator systems
 - Embedded system projects
 - Learning Arduino interfacing
 - Educational mini-projects
 - Digital input-output systems
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CONCLUSION

The Simple Calculator using Arduino UNO, 4x4 Keypad, and SSD1306 OLED Display was successfully implemented and tested. The system successfully performs basic arithmetic operations and displays the result

on the OLED screen. During the experiment I faces problem in mapping those keys and assign them their values. I overcome it by the open source resources given in web and arduino forum

REFERENCES

1. Arduino Keypad Library Documentation
<https://www.arduino.cc/reference/en/libraries/keypad/>
 2. SSD1306 OLED Display Library
https://github.com/adafruit/Adafruit_SSD1306
 3. Arduino UNO Official Documentation
<https://docs.arduino.cc/hardware/uno-rev3/>
 4. Adafruit GFX Graphics Library
<https://github.com/adafruit/Adafruit-GFX-Library>
 5. 4×4 Keypad Arduino Code, Pinout & Interfacing Complete Guide
<https://robosans.com/learn/embedded/arduino/4x4-keypad-arduino-code-pinout-interfacing-lcd/>
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